

```
A <- matrix(c(1,2,3,4), ncol = 2, nrow=2, byrow = TRUE)
det(A)
```

```
[1] -2
```

Inverse of Matrices

The inverse of a matrix is given by the function `solve()`

```
A <- matrix(c(1,2,3,4), ncol = 2, nrow=2, byrow = TRUE)
solve(A)
```

```
      [,1] [,2]
[1,] -2.0  1.0
[2,]  1.5 -0.5
```

Transpose of Matrices

The transpose of a matrix is given by the function `solve()`

```
A <- matrix(c(1,2,3,4), ncol = 2, nrow=2, byrow = TRUE)
t(A)
```

```
      [,1] [,2]
[1,]     1     3
[2,]     2     4
```

Plots in R using ggplot2 Package

In this chapter we see how to plot using the `ggplot2` package in R.

To install the package use the following syntax:

```
install.packages("ggplot2")
```

Components of plots

All the plots are composed of the following basics:

Data

Data that you want to visualise and a set of aesthetic mappings describing how variables in the data are mapped to aesthetic attributes that you can perceive.

Layers

Layers made up of geometric elements and statistical transformation. Geometric objects, `geoms` for short, represent what you actually see on the plot: points, lines, polygons, etc.

Scales

The scales map values in the data space to values in an aesthetic space, whether it be colour, or size, or shape. Scales draw a legend or axes, which provide an inverse mapping to make it possible to read the original data values from the plot.

Faceting

A faceting specification describes how to break up the data into subsets and how to display those subsets as small multiples. This is also known as conditioning or latticing/trellising.

Theme

A theme which controls the finer points of display, like the font size and background colour.

Getting Started in ggplot2

The goal of this section is to teach you how to produce useful graphics with `ggplot2` as quickly as possible.

We will use the inbuilt fuel economy data that is bundled with `ggplot2` package.

It includes information about the fuel economy of popular car models in 1999 and 2008.

You can have access to the data by loading `ggplot2`

```
library(ggplot2)
knitr::kable(head(mpg))
```

manufac- turer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	com- compact
audi	a4	1.8	1999	4	man- ual(m5)	f	21	29	p	com- compact
audi	a4	2.0	2008	4	man- ual(m6)	f	20	31	p	com- compact
audi	a4	2.0	2008	4	auto(av)	f	21	30	p	com- compact
audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	com- compact
audi	a4	2.8	1999	6	man- ual(m5)	f	18	26	p	com- compact

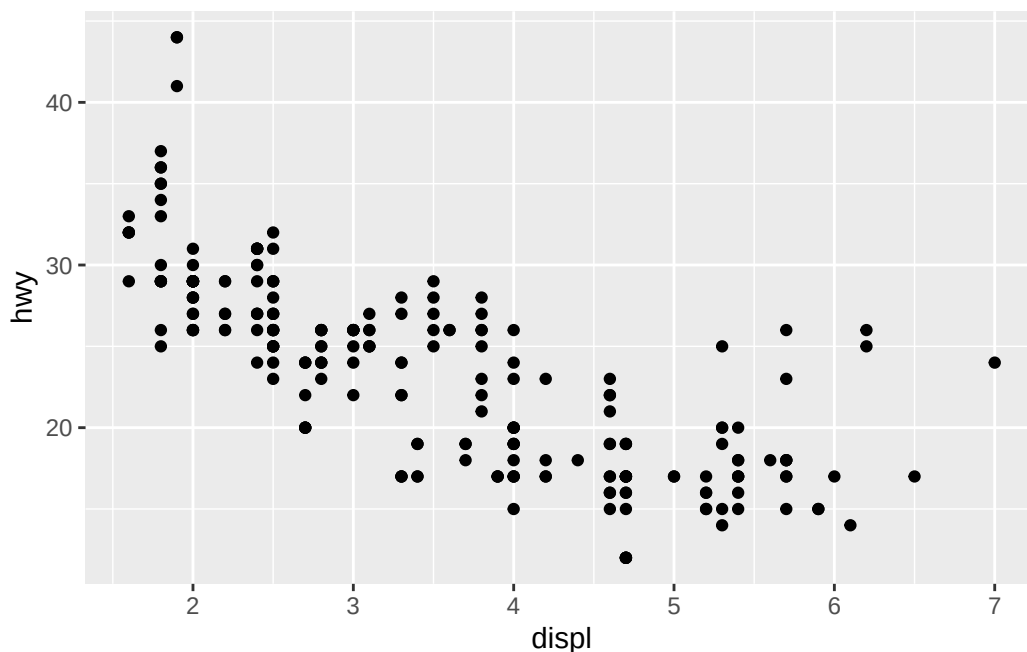
Key components

Every ggplot2 plot has three key components:

- **data** the data to be used to do plots
- **aesthetic mapping** A set of aesthetic mapping between variables in the data and visual properties
- At least one layer which describes how to render each observation. Layers are usually created with the **geom** function

Here gives a simple example

```
ggplot(mpg, aes(x = displ, y = hwy)) +  
  geom_point()
```



This produces a simple scatter plot defined by:

1. Data: `mpg`
2. Aesthetic mapping: `displ` mapped to x position, `hwy` mapped to y position
3. Layers: points

Pay attention to the structure of this function call: data and aesthetic mapping are supplied to `ggplot()` then layers are added on with `+`

Basic Plots

We explore the following basic plots in R and see how to plot using the `ggplot2()` package